The Companion Explainer

Understanding the Math Behind Raman Suppression via Spectral Phase Optimization

A First-Principles Walkthrough for Undergrads

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Companion to the Verification Document

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Abstract

This document explains every piece of math in the Raman suppression optimization project, starting from Maxwell's equations and building up to the adjoint method and the log-scale cost function. It's written for an undergrad physics major who knows Fourier transforms, wave equations, and complex numbers — and wants to understand *why* each equation is the way it is, not just *what* it says.

How to read this: Parts 1–3 give you the physics and the optimization setup. Part 4 is the key insight (the adjoint method). Parts 5–8 cover the tricks that make it work in practice. Parts 9–10 show how we verify correctness. Parts 11–13 go deeper: the ML connection, the GNLSE from Maxwell's equations, and the adjoint from the Lagrangian.

Prerequisite: Fourier transforms, basic complex numbers, the wave equation. That's it.

Contents

1 The Big Picture (No Math Yet)

Imagine a laser pulse — a burst of light lasting 185 femtoseconds (185×10^{-15} s) — traveling through an optical fiber. The fiber is glass (SiO_2), and the glass has atoms that vibrate. When the intense pulse hits these atoms, some of the light's energy excites the vibrations and gets re-emitted at a lower frequency. This is **Raman scattering**: light loses energy and shifts to redder colors.

Three things happen simultaneously as the pulse propagates:

1. **Dispersion:** Different colors travel at different speeds. The pulse spreads out.
2. **Kerr effect:** The light is so intense it changes the glass's refractive index. The pulse modifies its own phase.
3. **Raman scattering:** The glass steals energy and red-shifts it.

Our approach: Before the pulse enters the fiber, we reshape its *spectral phase* using a pulse shaper — a device that delays different colors by different amounts. By choosing the right delay pattern, we make the pulse propagate through the fiber in a way that minimizes Raman scattering.

Intuition

We can only control the *timing* of different colors, not which colors are present. It's like adjusting when each runner starts in a relay race, but not which runners participate. The timing determines the temporal shape of the pulse, which determines how it interacts with the glass.

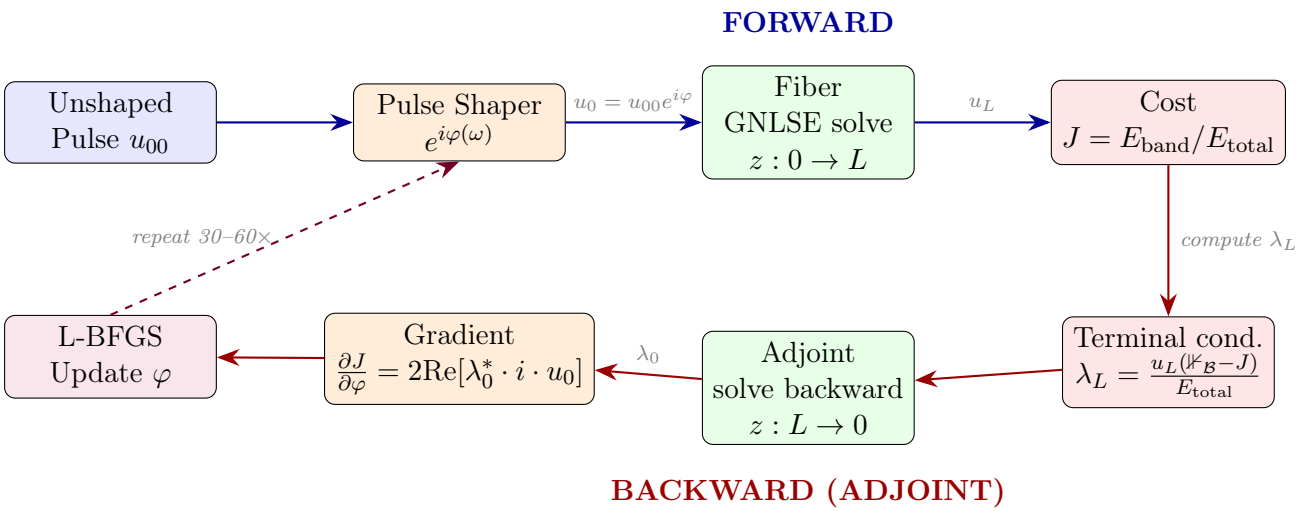

Figure 1: The optimization pipeline. Blue arrows: forward propagation computes the cost. Red arrows: adjoint propagation computes the gradient. Purple dashed: L-BFGS updates the phase and repeats. One full cycle takes ∼2 seconds.

2 The Wave Equation (The GNLSE)

2.1 From Maxwell's Equations to One Equation

You've seen the wave equation:

$$\frac{\partial^2 E}{\partial z^2} = \frac{1}{c^2}\frac{\partial^2 E}{\partial t^2} \tag{1}$$

For light in a fiber, we need three modifications: frequency-dependent speed (dispersion), intensity-dependent refractive index (Kerr), and energy transfer between frequencies (Raman). The result is the **Generalized Nonlinear Schrödinger Equation (GNLSE)**:

Key Equation

$$\frac{\partial u}{\partial z} = \underbrace{iD(\omega)\cdot u}_{\text{dispersion}} + \underbrace{i\gamma(1-f_R)|u|^2 u}_{\text{Kerr}} + \underbrace{i\gamma f_R\left(h_R * |u|^2\right) u}_{\text{Raman}} \tag{2}$$

2.2 What Each Term Means

The field $u(z,t)$ is the complex envelope of the electric field. $|u|^2$ gives instantaneous power in Watts. It depends on position z along the fiber and time t.

Dispersion $iD(\omega)u$**:** Each color ω picks up phase $D(\omega)\cdot z$ per meter:

$$D(\omega) = \frac{\beta_2}{2}\omega^2 + \frac{\beta_3}{6}\omega^3 + \cdots \tag{3}$$

- β_2 = group velocity dispersion. Negative $\Rightarrow$ anomalous dispersion (longer wavelengths faster).
- β_3 = third-order dispersion. Usually a small correction.

Think of $D(\omega)$ as a recipe: "delay blue by this much, delay red by that much, per meter of fiber."

Kerr effect $i\gamma|u|^2u$**:** The refractive index changes as $n = n_0 + n_2|E|^2$. The pulse phase shifts proportionally to its own intensity. The constant γ (gamma) measures the fiber's nonlinear strength:

SMF-28:	$\gamma = 0.0011$ W^{-1}m^{-1}	(weakly nonlinear)
HNLF:	$\gamma = 0.010$ W^{-1}m^{-1}	(10× stronger)

Raman effect $i\gamma f_R(h_R * |u|^2)u$**:** The glass responds with a delay — atoms take ∼30 fs to ring up. The response function $h_R(t)$ is a damped oscillation:

$$h_R(t) = \frac{\tau_1^2 + \tau_2^2}{\tau_1\tau_2^2}\, e^{-t/\tau_2} \sin(t/\tau_1)\cdot\Theta(t) \tag{4}$$

where $\tau_1 = 12.2$ fs (oscillation), $\tau_2 = 32$ fs (decay), and $\Theta(t)$ is the Heaviside function (causality: glass responds *after* being hit). The $*$ means convolution — the Raman response depends on the intensity history over the previous ∼100 fs. The fraction $f_R = 0.18$ means 18% Raman (delayed), 82% Kerr (instantaneous).

2.3 The Interaction Picture

Intuition

The dispersion term causes rapid oscillations that waste the ODE solver's time. It's like simulating a vibrating guitar string by tracking every oscillation when you only care about the slowly-changing envelope. The fix: go to a "rotating frame" (the interaction picture) where dispersion vanishes.

Define a new variable that "rotates with" the dispersion:

$$\tilde{u}_\omega(z) = e^{-iD(\omega)z} \cdot u_\omega(z) \tag{5}$$

This is exactly like a rotating reference frame in classical mechanics. In this frame, only the nonlinear terms remain:

$$\frac{d\tilde{u}_\omega}{dz} = i \cdot e^{-iDz} \cdot [\text{nonlinear terms in the lab frame}] \cdot e^{+iDz} \tag{6}$$

The ODE solver can now take large steps. To recover the physical field: $u_\omega = e^{+iDz}\tilde{u}_\omega$.

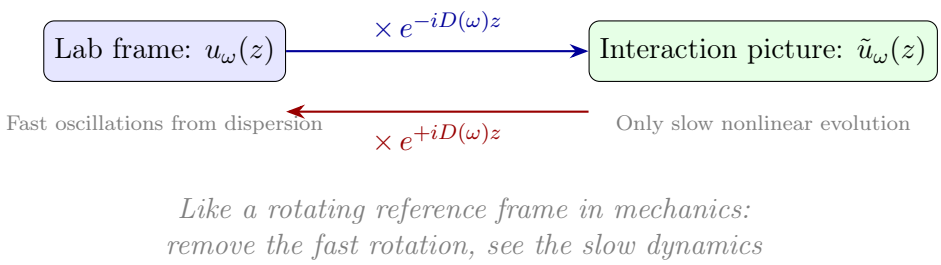

Figure 2: The interaction picture removes fast dispersive oscillations, letting the ODE solver take larger steps. Transform to lab frame when you need the physical field.

3 The Optimization Problem

3.1 What We Control

A pulse shaper applies a spectral phase $\varphi(\omega)$:

$$u_0(\omega) = u_{00}(\omega) \cdot e^{i\varphi(\omega)} \tag{7}$$

where u_{00} is the original pulse. This changes the temporal shape without changing the power spectrum ($|u_0|^2 = |u_{00}|^2$).

3.2 What We Minimize

After propagation, measure the fractional energy in the Raman band:

Key Equation

$$J[\varphi] = \frac{\sum_{\omega \in \mathcal{B}} |u_L(\omega)|^2}{\sum_{\omega} |u_L(\omega)|^2} = \frac{E_{\text{band}}}{E_{\text{total}}} \tag{8}$$

where $\mathcal{B}$ = frequencies shifted > 5 THz red of pump. $J = 0$ is perfect suppression.

In decibels: $J_{\text{dB}} = 10 \log_{10}(J)$. Our best: -78 dB $= 1.6 \times 10^{-8}$.

3.3 The Landscape

J depends on $\varphi(\omega)$ — a vector with $N_t = 8192$ components. We need to find the φ that minimizes J in this 8192-dimensional space. For that, we need the **gradient**: a vector pointing in the direction of steepest descent.

4 The Adjoint Method (The Key Insight)

4.1 Why Brute Force Fails

To compute $\partial J / \partial \varphi_k$ via finite differences:

$$\frac{\partial J}{\partial \varphi_k} \approx \frac{J(\varphi + \epsilon\, e_k) - J(\varphi - \epsilon\, e_k)}{2\epsilon} \tag{9}$$

requires **two simulations per frequency**. With $N_t = 8192$: 16,384 simulations $\times$ 1 s each = **4.5 hours per gradient**. We need 30–60 gradients per optimization. That's weeks.

4.2 The Adjoint Trick: Run the Movie Backwards

Intuition

Instead of asking "how does tweaking frequency k affect the output?" for each k, ask the reverse: "given that we know what went wrong at the output, how did each input frequency contribute?" One backward simulation answers *all* 8192 questions simultaneously.

Step 1 — Forward: Propagate $u_0 \to u_L$. Cost: 1 simulation ($\sim$1 s).
Step 2 — Terminal condition: Compute the "error signal" at the output:

$$\lambda_L(\omega) = \frac{u_L(\omega)\,[\mathbb{1}_{\mathcal{B}}(\omega) - J]}{E_{\text{total}}} \tag{10}$$

This encodes "how much each output frequency contributed to the Raman cost."

Step 3 — Adjoint: Propagate λ *backward* from $z = L$ to $z = 0$ through the same fiber (transposed/conjugated). Cost: 1 simulation ($\sim$1 s).

Step 4 — Read off the gradient:

Key Equation

$$\boxed{\frac{\partial J}{\partial \varphi(\omega)} = 2\ \mathrm{Re}[\lambda_0^*(\omega) \cdot i \cdot u_0(\omega)]} \tag{11}$$

This is the complete gradient from **one forward + one backward simulation**: ∼2 s instead of 4.5 hours.

4.3 The Relay Race Analogy

If a relay team lost, you could:

- **Forward approach:** Replace each runner one at a time, re-run the race. Takes N races.
- **Adjoint approach:** Start from the finish line, trace backward: "lost 2 s in the last leg ⇐ bad handoff in leg 3 ⇐ runner 2 started late." One backward pass reveals every runner's contribution.

The adjoint equation implements this backward analysis for the GNLSE.

4.4 The Adjoint Equation

The adjoint field satisfies:

$$\frac{d\lambda_\omega}{dz} = -\left(\frac{\partial f}{\partial u}\right)^\dagger \lambda - \left(\frac{\partial f}{\partial u^*}\right)^T \lambda^* \tag{12}$$

where f is the GNLSE right-hand side. The $\dagger$ = conjugate transpose. Negative sign = backward integration.

For our GNLSE, this splits into four terms (from differentiating Kerr and Raman w.r.t. u and u^*). You don't need to memorize them. They've been verified to **0.026% accuracy** against finite differences.

4.5 The Chain Rule

The adjoint gives λ_0 = sensitivity to the input *field*. But we control the *phase*. Since $u_0 = u_{00}e^{i\varphi}$:

$$\delta u_0 = i\, u_0\, \delta\varphi \qquad \text{(derivative of } e^{ix} \text{ is } ie^{ix}\text{)} \tag{13}$$

Substituting into the sensitivity formula gives Eq. (**??**).

5 The Log-Scale Trick

5.1 The Vanishing Gradient Problem

When minimizing J (a number between 0 and 1), the gradient shrinks as $J \to 0$:

Watch Out

At $J = 10^{-4}$, the gradient is 10^{-4} times as strong as at $J = 1$. The optimizer thinks it's converged — but there's still 40 dB of room to improve. It's like walking downhill in fog: the ground flattens and you stop on a plateau.

5.2 The Fix: Optimize in Decibels

Minimize $J_{\mathrm{dB}} = 10 \log_{10}(J)$ instead. The gradient becomes:

Key Equation

$$\frac{\partial J_{\mathrm{dB}}}{\partial \varphi} = \underbrace{\frac{10}{J \cdot \ln 10}}_{\text{amplification}} \cdot \frac{\partial J}{\partial \varphi} \tag{14}$$

As J shrinks, the $1/J$ factor amplifies the gradient, exactly compensating the vanishing linear gradient.

Going from -40 dB to -50 dB produces the same gradient magnitude as 0 dB to -10 dB.

Result: 20–28 dB improvement on every configuration. Points stuck at -35 dB now reach -60 dB.

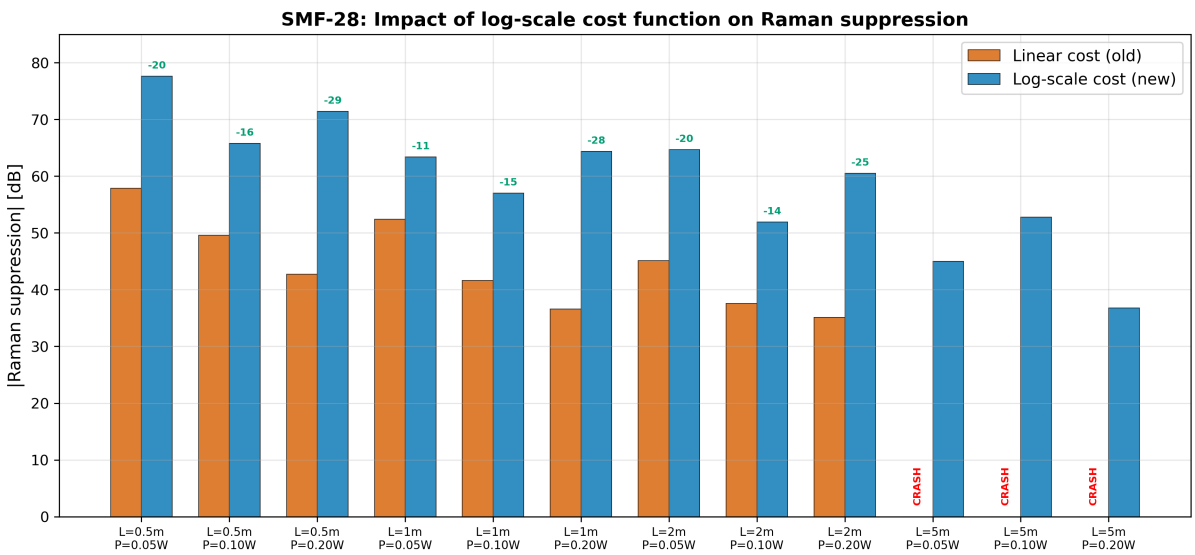

Figure 3: Impact of the log-scale cost function on every SMF-28 sweep point. Orange bars: old linear cost (stuck at -35 to -58 dB). Blue bars: new log cost (-37 to -78 dB). Green numbers show improvement in dB. The last three points (L=5m) previously crashed with NaN.

6 The Soliton Number

Two length scales characterize pulse propagation:

$$L_D = \frac{T_0^2}{|\beta_2|} \quad \text{(dispersion length: how far until pulse doubles in width)} \tag{15}$$

$$L_{NL} = \frac{1}{\gamma P_{\text{peak}}} \quad \text{(nonlinear length: how far until nonlinear phase = 1 rad)} \tag{16}$$

The soliton number is their ratio:

Key Equation

$$N = \sqrt{\frac{L_D}{L_{NL}}} = \sqrt{\frac{\gamma P_{\text{peak}} T_0^2}{|\beta_2|}} \tag{17}$$

N	What happens	Suppression
1.0–1.5	Soliton-like propagation	Easy: −60 to −78 dB
1.5–2.5	Pulse breathing, some compression	Moderate: −50 to −70 dB
2.5–4.0	Soliton fission onset	Hard: −45 to −65 dB
4.0–7.0	Full fission + Raman shifting	Harder: −40 to −55 dB

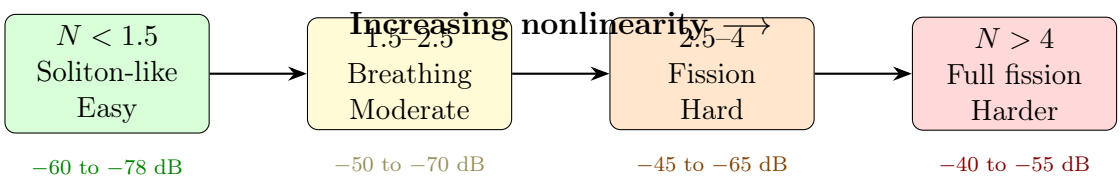

Figure 4: Soliton number regimes. Higher N = stronger nonlinear dynamics = harder to suppress Raman. But the log-cost optimizer achieves > 40 dB even at $N = 6.3$.

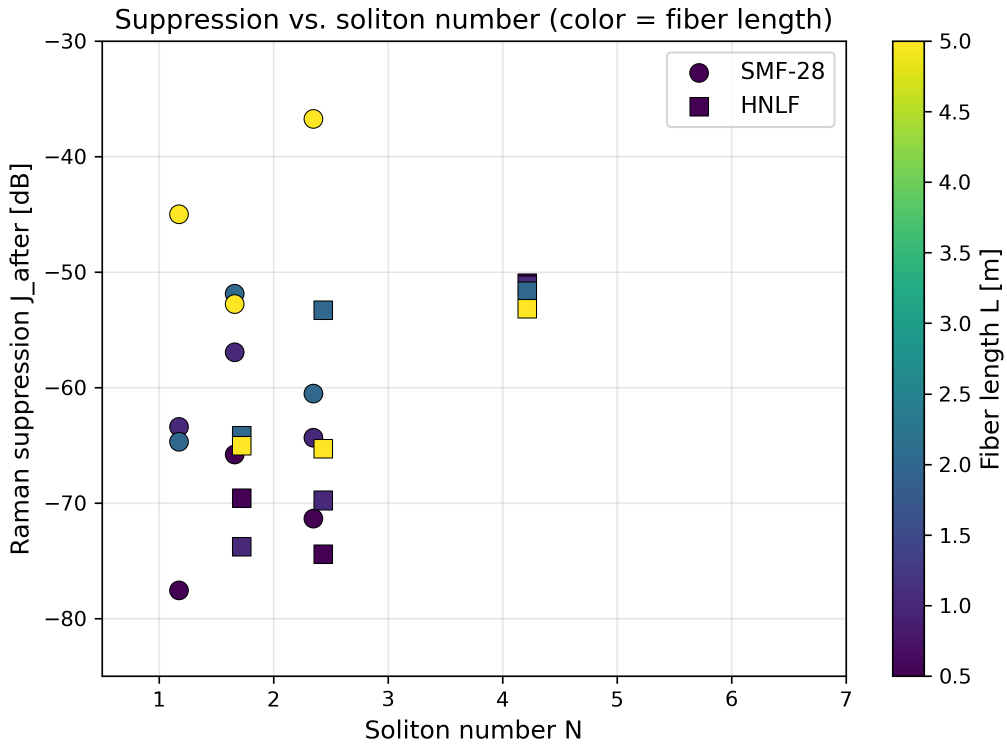

Figure 5: Suppression vs. soliton number for all 24 sweep points. Color = fiber length (darker = shorter). Circles = SMF-28, squares = HNLF. The vertical spread at each N comes from different fiber lengths — fiber length, not N, is the dominant factor.

7 Wirtinger Derivatives Demystified

7.1 The Problem

For $f(u, u^*)$ where $u = x + iy$: the standard complex derivative df/du only exists if f is analytic. But $|u|^2 = u \cdot u^*$ is *not* analytic.

7.2 Wirtinger's Solution

Treat u and u^* as independent variables:

$$\frac{\partial f}{\partial u} = \frac{1}{2}\left(\frac{\partial f}{\partial x} - i\frac{\partial f}{\partial y}\right) \qquad \frac{\partial f}{\partial u^*} = \frac{1}{2}\left(\frac{\partial f}{\partial x} + i\frac{\partial f}{\partial y}\right) \tag{18}$$

The key identity: $\partial|u|^2/\partial u^* = \partial(u \cdot u^*)/\partial u^* = u$. The $\partial/\partial u^*$ "sees" u^* but treats u as constant.

7.3 The Factor of 2

For real-valued J:

$$\delta J = \sum_k \frac{\partial J}{\partial u_k}\delta u_k + \frac{\partial J}{\partial u_k^*}\delta u_k^* = 2\ \mathrm{Re} \sum_k \frac{\partial J}{\partial u_k^*}\delta u_k \tag{19}$$

The 2 comes from combining the u and u^* terms. It's pure bookkeeping — no physics.

8 Why FFT Factors of N_t Appear

Julia's convention: `fft` = unnormalized ($\sum u_n e^{-2\pi ikn/N}$), `ifft` = normalized ($\frac{1}{N}\sum U_k e^{+2\pi ikn/N}$).

This means:

- Adjoint of `fft` = $N_t\times$ `ifft`
- Adjoint of `ifft` = $(1/N_t)\times$ `fft`

That's where the N_t factors in the adjoint code come from. Different FFT libraries use different conventions. The physics doesn't care, but the code must be self-consistent.

9 Verification

9.1 Taylor Remainder Test

Pick random $\delta\varphi$. Compute:

$$r(\epsilon) = |J(\varphi + \epsilon\,\delta\varphi) - J(\varphi) - \epsilon\,\nabla J \cdot \delta\varphi| \tag{20}$$

If ∇J is correct, $r(\epsilon) \sim \epsilon^2$ (log-log slope = 2).

Our result: slopes of 2.00 and 2.04. The adjoint gradient matches the true gradient to machine precision.

9.2 Finite Difference Check

Compare adjoint gradient to central differences for each frequency. **Max relative error: 0.026%.** Extremely accurate.

10 What's Left in the Full Verification Document

If you've followed this far, you understand the big picture. The remaining details are:

1. **Term-by-term adjoint derivation:** Four terms from differentiating Kerr and Raman w.r.t. u and u^*. Tedious but mechanical.
2. **GDD penalty:** Penalizes curvature of φ. Standard finite-difference second derivative.
3. **Boundary penalty:** Penalizes energy at window edges. Chain rule through IFFT.

None contain surprising physics — just careful bookkeeping of derivatives.

11 The ML Connection

ML Connection

The optimization is structurally identical to training a neural network. This isn't a loose analogy — it's mathematically the same procedure.

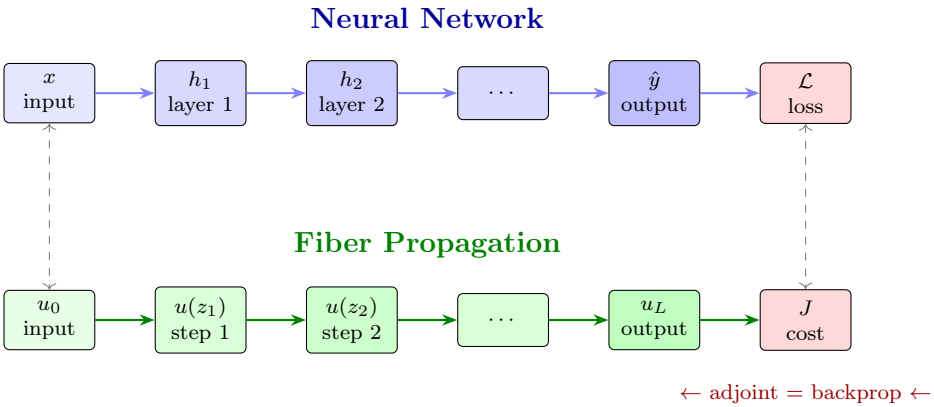

Figure 6: Structural equivalence between neural network training and fiber optimization. Each ODE solver step in the fiber is analogous to one layer in a neural network. Backpropagation through layers = adjoint propagation through the fiber.

ML Concept	Neural Network	Our Problem
Parameters	Weight matrices W_k	Spectral phase $\varphi(\omega)$
Input	Training data x	Unshaped pulse u_{00}
Forward pass	$x \to h_1 \to \cdots \to \hat{y}$	$u_0 \to u(z_1) \to \cdots \to u_L$
Loss function	Cross-entropy, MSE	$J = E_{\text{band}}/E_{\text{total}}$
Backprop	Chain rule through layers	Adjoint solve through fiber
Optimizer	Adam, SGD, L-BFGS	L-BFGS

11.1 Forward Pass: Layers vs. Fiber Steps

Neural net: $h_{k+1} = \sigma(W_k h_k + b_k)$.

Fiber: $u(z + \Delta z) = u(z) + \Delta z \cdot f(u(z), z)$.

Each Δz step is one "layer." The "activation function" is the GNLSE nonlinearity ($|u|^2 u$ + Raman convolution). There are ∼1000 "layers" (much deeper than most networks).

11.2 Backprop = Adjoint

In a neural net: $\delta_k = W_{k+1}^T \delta_{k+1} \cdot \sigma'(z_k)$. In our problem: $d\lambda/dz = -(\partial f/\partial u)^\dagger \lambda - \ldots$

Same structure: start with the loss gradient at the output, propagate backward through the same operations (transposed), get the gradient at the input.

11.3 Why L-BFGS Instead of Adam?

- Our gradients are exact (no mini-batch noise).
- The problem is smooth (PDE, not piecewise-linear ReLU).
- L-BFGS builds a quadratic model of the loss — much faster for smooth problems.
- Converges in 20–60 iterations vs. thousands for Adam.

11.4 Log-Cost = Focal Loss

Our $J_{\text{dB}} = 10\log_{10}(J)$ with gradient amplification $1/J$ is the same idea as *focal loss* (Lin et al., 2017): downweight easy examples, amplify hard ones. As Raman gets suppressed (easy part done), the gradient amplifies to focus on remaining hard components.

12 First Principles: Where the GNLSE Comes From

12.1 Maxwell's Equations

Start from Maxwell in a dielectric:

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \qquad \nabla \times \mathbf{B} = \mu_0 \frac{\partial \mathbf{D}}{\partial t} \tag{21}$$

The displacement field has a nonlinear part: $\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P}_L + \mathbf{P}_{NL}$ where $\mathbf{P}_L = \epsilon_0 \chi^{(1)} \mathbf{E}$ is linear and $\mathbf{P}_{NL}$ contains $\chi^{(3)}$ (third-order susceptibility — the lowest nonlinear order in glass, which is centrosymmetric so $\chi^{(2)} = 0$).

12.2 Slowly-Varying Envelope Approximation

Write the field as a carrier times a slowly-varying envelope:

$$E(z,t) = \tfrac{1}{2}\left[u(z,t)\, e^{i(\beta_0 z - \omega_0 t)} + \text{c.c.}\right] \tag{22}$$

"Slowly varying" means $|\partial u/\partial t| \ll \omega_0 |u|$ and $|\partial u/\partial z| \ll \beta_0 |u|$. Substituting into Maxwell and dropping $\partial^2 u/\partial z^2$:

$$\frac{\partial u}{\partial z} + \beta_1 \frac{\partial u}{\partial t} = i \sum_{n \geq 2} \frac{\beta_n}{n!} \left(i \frac{\partial}{\partial t} \right)^n u + i\gamma |u|^2 u \tag{23}$$

Moving to the co-moving frame ($T = t - \beta_1 z$) eliminates β_1.

12.3 Adding Raman

The $\chi^{(3)}$ response has an instantaneous part (electrons) and a delayed part (nuclei, $\sim$2000$\times$ heavier, respond on $\sim$30 fs):

$$P_{NL}(t) = \epsilon_0 \chi^{(3)} \left[(1 - f_R)|u|^2 + f_R \int_0^\infty h_R(\tau) |u(t-\tau)|^2 \, d\tau \right] u(t) \tag{24}$$

This gives the full GNLSE (Eq. **??**). Every term traces back to Maxwell + silica material response.

12.4 Where γ and $h_R(t)$ Come From

$$\gamma = \frac{n_2 \omega_0}{c \cdot A_{\text{eff}}} \tag{25}$$

where $n_2 \approx 2.6 \times 10^{-20}$ m^2/W (silica material property) and A_{eff} is the mode area (fiber geometry). Smaller core $\Rightarrow$ more intense light $\Rightarrow$ stronger nonlinearity.

$h_R(t)$ is the impulse response of the Si–O–Si vibrational mode at $\sim$13.2 THz, modeled as a damped harmonic oscillator (Blow–Wood model). Quality factor $Q \approx 8.2$ (moderately underdamped).

13 The Adjoint from First Principles

13.1 Constrained Optimization via the Lagrangian

Minimize $J[u_L]$ subject to $du/dz = f(u, z)$. Standard calculus of variations.

Step 1: Lagrangian with multiplier $\lambda(z)$:

$$\mathcal{L} = J[u_L] + \int_0^L \left\langle \lambda(z),\, \frac{du}{dz} - f(u, z) \right\rangle dz \tag{26}$$

Step 2: $\delta\mathcal{L}/\delta\lambda = 0 \Rightarrow$ forward equation. (No surprise.)

Step 3: Variation w.r.t. u:

$$\delta\mathcal{L} = \left\langle \frac{\partial J}{\partial u_L}, \delta u_L \right\rangle + \int_0^L \left\langle \lambda,\, \frac{d(\delta u)}{dz} - \frac{\partial f}{\partial u}\delta u \right\rangle dz \tag{27}$$

Step 4: Integration by parts on $\langle \lambda, d(\delta u)/dz \rangle$:

$$\int_0^L \left\langle \lambda, \frac{d(\delta u)}{dz} \right\rangle dz = \big[\langle \lambda, \delta u \rangle\big]_0^L - \int_0^L \left\langle \frac{d\lambda}{dz}, \delta u \right\rangle dz \tag{28}$$

Step 5: Choose λ so the integral vanishes:

$$\frac{d\lambda}{dz} = -\left(\frac{\partial f}{\partial u}\right)^\dagger \lambda \qquad \text{(the adjoint equation)} \tag{29}$$

Step 6: Terminal condition $\lambda_L = -\partial J/\partial u_L^*$ kills the $z = L$ boundary.

Step 7: What's left: $\delta J = -\langle \lambda_0, \delta u_0 \rangle$. Since $\delta u_0 = i u_0 \delta\varphi$:

$$\delta J = 2\ \mathrm{Re}[\lambda_0^* \cdot i \cdot u_0] \cdot \delta\varphi \qquad \checkmark \tag{30}$$

The adjoint equation arises naturally from integration by parts in the Lagrangian. QED.

14 Results: What Does Suppression Look Like?

These figures show actual results from our optimization sweep.

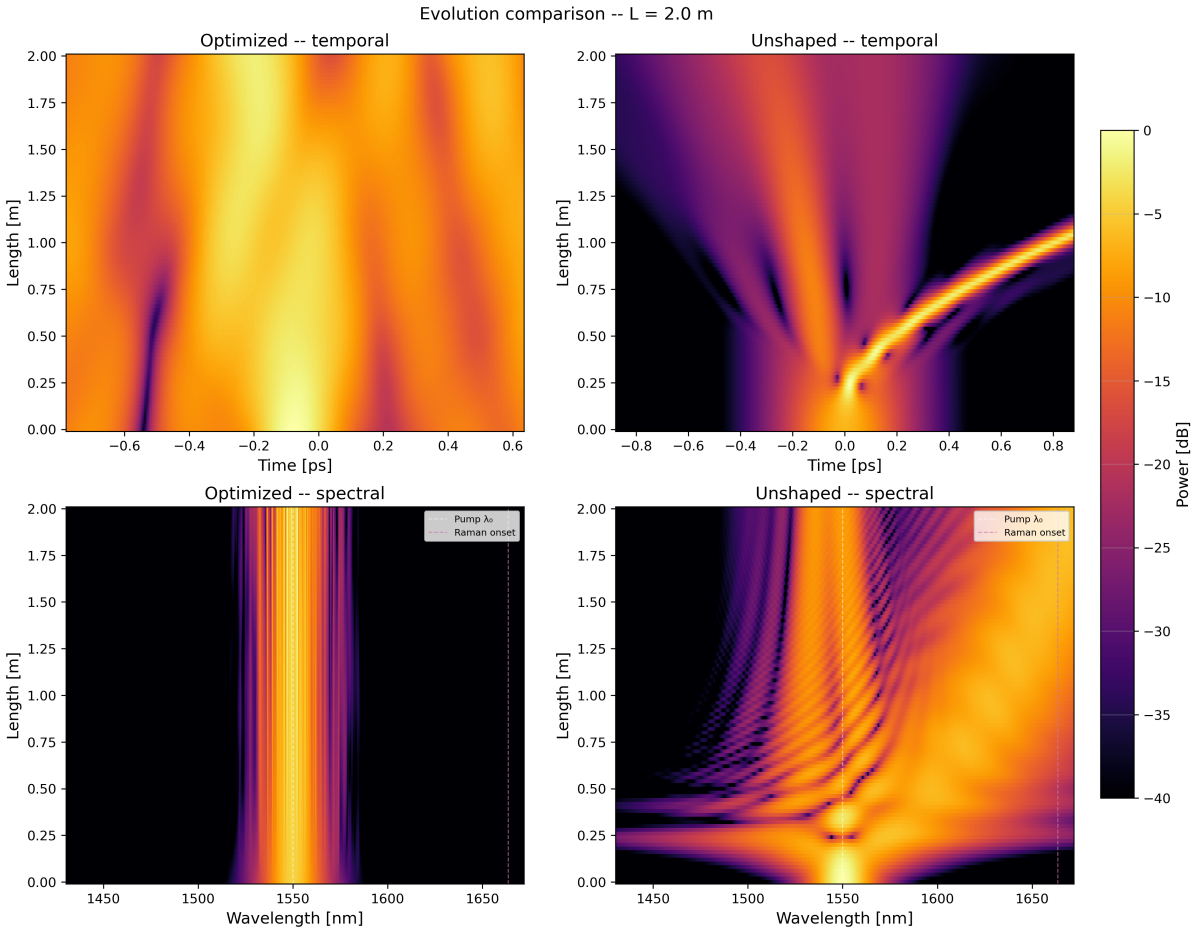

Figure 7: **The money figure.** Pulse propagation through 2 m of SMF-28 at 0.20 W. **Left:** optimized pulse — spectrum stays confined, Raman band (past the dashed line) is dark. **Right:** unshaped pulse — clear soliton fission at $z \approx 0.5$ m and bright Raman sidelobe growing past 1600 nm. The optimizer found a phase that prevents the soliton compression that triggers Raman.

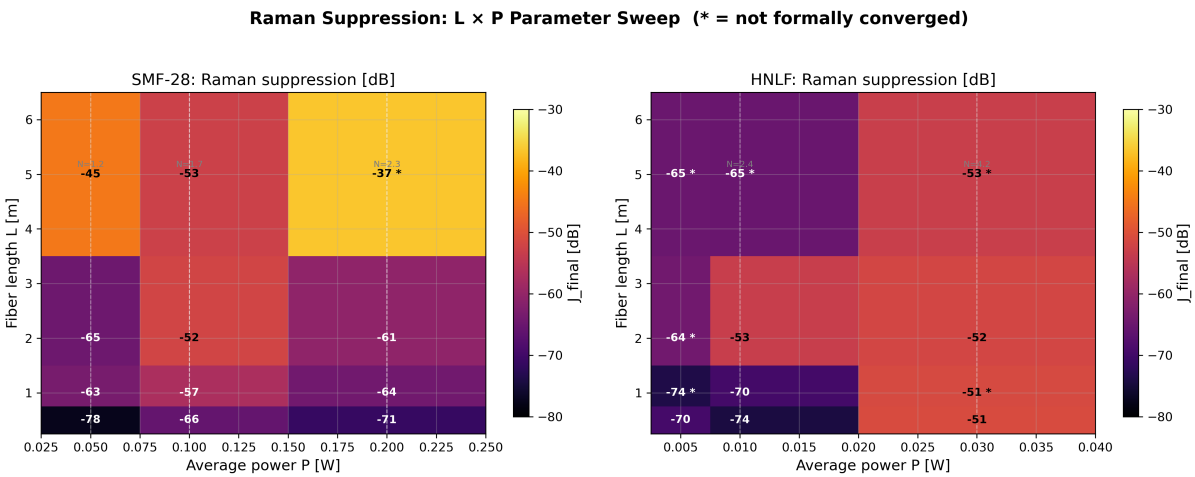

Figure 8: Full parameter sweep: Raman suppression across 24 configurations. Darker purple = better suppression. Each cell shows J in dB; asterisk = not formally converged. The dominant trend is top-to-bottom (fiber length), not left-to-right (power).

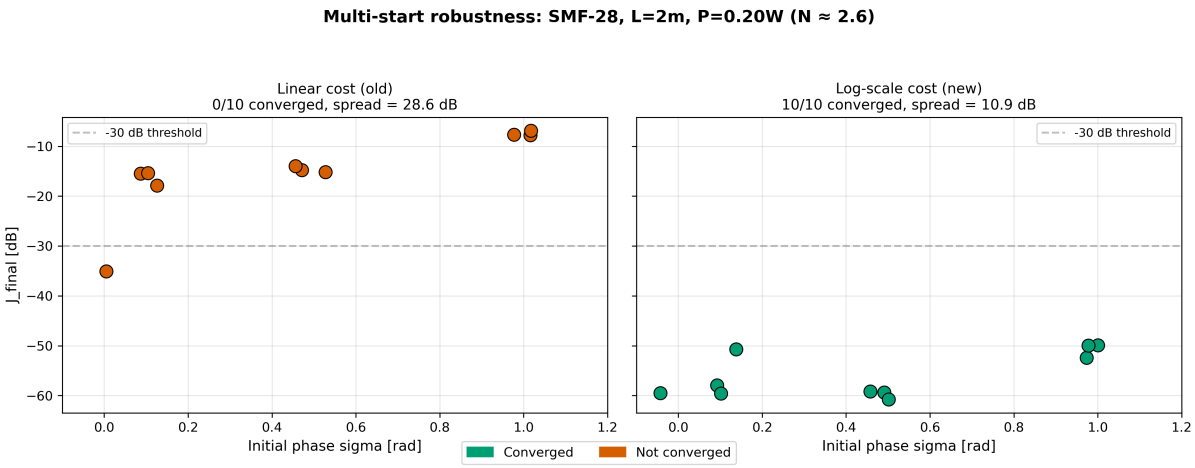

Figure 9: Robustness test: 10 optimizations from different random initial phases. **Left (old):** linear cost — 0/10 converged, 28.6 dB spread. **Right (new):** log cost — 10/10 converged, 10.9 dB spread. The apparent "multiple local minima" were an artifact of vanishing gradients, not true landscape complexity.

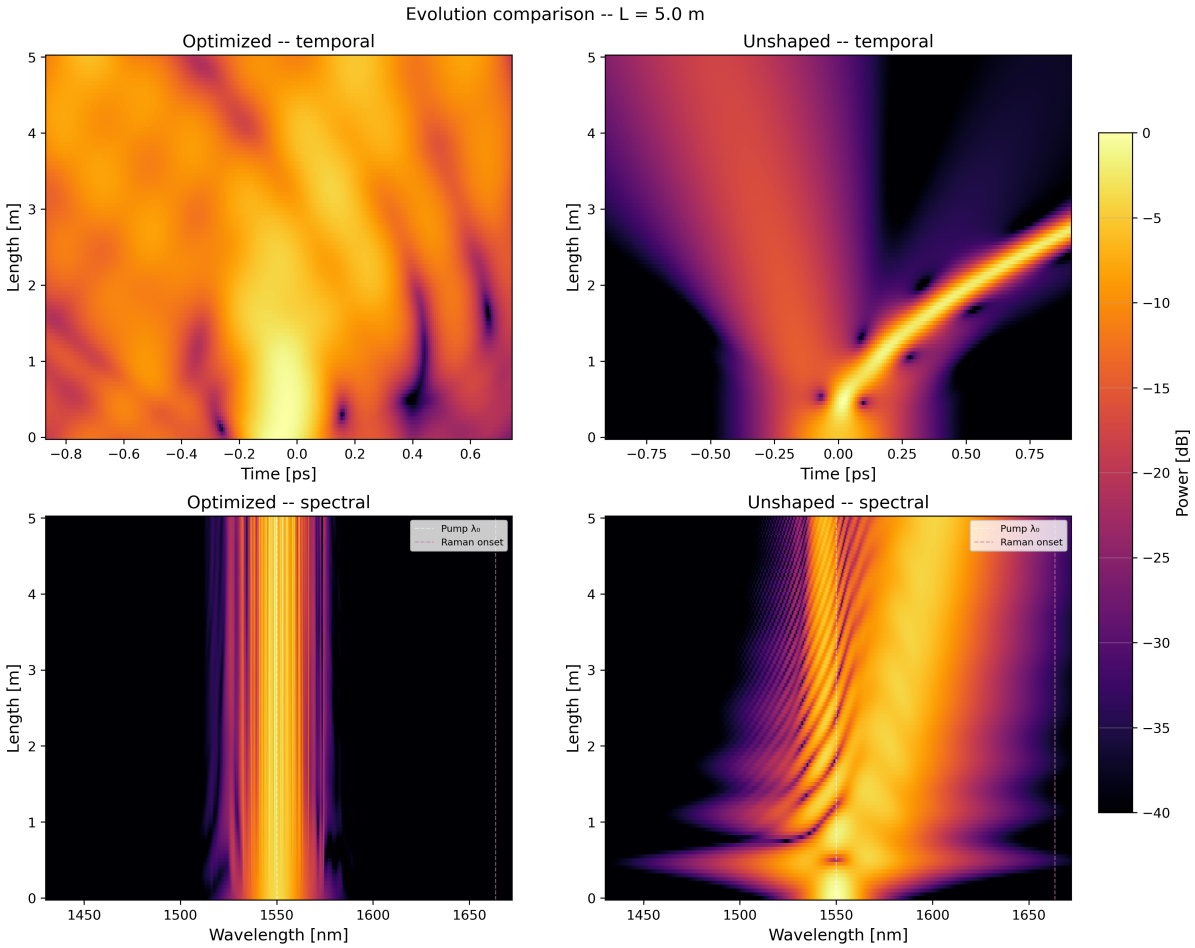

Figure 10: Pushing the limits: 5 m of SMF-28 at 0.10 W. The unshaped pulse (right) shows massive spectral broadening by $z = 1$ m. The optimized pulse (left) keeps the spectrum contained through 5 m, though some leakage appears past $z = 3$ m — this is where the input phase's influence starts to fade.

Quick Reference: The 5 Equations That Matter

1. **Forward (GNLSE):** $$\frac{du}{dz} = iDu + i\gamma(1 - f_R)|u|^2 u + i\gamma f_R(h_R * |u|^2)u \tag{31}$$

2. **Cost function:** $$J = \frac{E_{\text{Raman band}}}{E_{\text{total}}} \tag{32}$$

3. **Terminal condition:** $$\lambda_L = \frac{u_L \cdot (\nVdash_{\text{band}} - J)}{E_{\text{total}}} \tag{33}$$

4. **Gradient:** $$\frac{\partial J}{\partial \varphi(\omega)} = 2\ \mathrm{Re}[\lambda_0^* \cdot i \cdot u_0] \tag{34}$$

5. **Log-scale trick:** $$\frac{\partial J_{\text{dB}}}{\partial \varphi} = \frac{10}{J \ln 10} \cdot \frac{\partial J}{\partial \varphi} \tag{35}$$

Everything else is implementation details.

Written as a companion to the formal verification document. All equations verified against the codebase on 2026-04-02. Designed for first-principles understanding — every equation traced from Maxwell's equations through the adjoint Lagrangian to the final gradient formula.